Experimenting in the Bedroom

Designing better sex toys for the FTM transgender and AFAB non-binary community.

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Acknowledgements

This senior thesis represents the culmination of my design education and personal growth at the Rensselaer Polytechnic Institute. As a designer and a transgender individual, I have poured everything I've learned and dreamed about into this project, to change the world for the better, even if just by a tiny bit.

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Introduction

Regardless of sexuality or gender identity, pleasure and sexual satisfaction is key to the well-being of many people. While this particular problem space has become less stigmatized, there is still a great amount of taboo surrounding both transgender bodies and sexual pleasure. As society begins to unravel this black box, the needs of gender minorities, such as women and queer folks continue to be sidelined. The majority of sex toys and literature regarding sex is created by and aimed at a straight, cisgender audience. As a result, many toys used by transgender individuals tend to be heavily adapted or modified versions of unsuitable products. A number of dysphoria interventions (binders, packers, etc.) are only addressed by small trans-owned businesses, simply because the larger market doesn't take our voices into account.

For transgender individuals in particular, the development of well-designed sex toys also presents an avenue for a type of medical intervention that is less invasive and costly than surgery, but more effective than simple prosthetics that are currently on the market.

This project will consolidate knowledge, experience, and unmet needs from the greater community of queer and gender non-conforming individuals through my social networks. As a result, a core element of the initial ideation and feedback gathering involved the distribution of an anonymous survey to capture the experiences of transmasculine individuals. It should also be stated that, due to the short timeframe of this project, I have designed for an FTM transmasculine audience, as that's also where my knowledge is situated. Additionally, foraying into designing MTF transfeminine products raises a bit of personal dysphoria—which makes sense, given that the MTF gender vector is directly opposite my own.

Terms List

The following are words and definitions that will be used throughout this project and paper. Some of these are the standard accepted terminology within the queer community as a whole, but other types of shorthand or slang may exist. While other terms may be synonymous, these specific words were chosen to account for dysphoria and to streamline discussion.

Anatomical Terms

- Dildo - This term refers to a type of phallic sex toy, used for penetrative stimulation of oneself or a partner.
- Penetration - Stimulating oneself or a partner with the insertion of toys, hands, or genitalia.
- Masturbation/Self-pleasure
- Oral sex
- Hand stimulation/hand stuff
- Giving/receiving (pleasure)

Gender & Gender Vectors

- Cisgender - An individual whose gender identity aligns with the one typically associated with the gender assigned to them at birth.
- Transgender - An individual whose gender identity that does not necessarily match their assigned gender at birth. A transgender person does not have to pursue options for transition in order to identify as such.
- Dysphoria - A state of discomfort and anxiety, caused by a jarring disconnect between one's experiences and their gender. Directly opposite of "gender euphoria", which is experienced when one's gender identity is affirmed. Not all transgender individuals experience this.
- Transitioning - This term encompasses doing so socially, legally, or medically.
- Female-to-Male (FTM) - An individual who was assigned or designated female at birth. Falls under the transgender umbrella.
- Male-to-Female (MTF) - An individual who was assigned or designated male at birth. Falls under the transgender umbrella.
- Non-binary (NB) - This term refers to someone who identifies outside the binary of male and female, regardless of assigned gender at birth. Falls under the transgender umbrella.
- Top/Bottom Surgery - These are forms of gender-affirming surgeries (GAS) that typically aim to permanently alter the physiology of a transgender individual's body to more closely resemble features associated with the gender they identify

with. Bottom surgery refers to any such surgery that addresses the function and appearance of reproductive organs, while top surgery focuses on the appearance of one's chest.

Literature Review

The section will explore how dysphoria is currently addressed, both through surgical and non-surgical interventions, societal taboos surrounding discussions of sexual pleasure, and sex toys as an alternative intervention point for dysphoria. While a great deal of research on sexual pleasure already exists, the majority of this literature is centered around a cisgender, heterosexual audience. One of the goals of my project has been to highlight the experiences of transgender individuals, to identify unmet problem spaces for research, and to better address those niches. As a result, this literature review seeks to weave together a narrative regarding the sexual experiences of transmasculine individuals and the ways this can be improved, using both social science and design literature.

Gender & Sexuality

The relationship one has to their own gender and sexuality is an incredibly intimate and personal topic, which can be different for everyone. For transgender individuals in particular, this relationship may develop and change throughout the course of their lives. Regardless, it is well-established that access to affirming body modifications (both permanent and non-permanent) positively impacts a person's sense of self and alleviates dysphoria (Schilt and Windsor). By addressing these needs, psychological well-being can be increased for transgender indviduals, just as sexual satisfaction correlates to body satisfaction in cisgender individuals. For those of us who desire sexual pleasure, affirming the mental models of how we want to give and receive pleasure is a key element to consider (Nikkelen and Kreukels).

Current Interventions for Transmasculine Individuals

While not every transgender person experiences dysphoria, a large number of us pursue medical interventions such as hormone replacement therapy (HRT) or surgery. While HRT is widely pursued and has satisfactory outcomes, it’s often not enough (Davis and Meier). As a result, a great deal of transmasculine individuals will seek gender affirming surgeries (GAS). The two main forms of the surgeries are colloquially known as “top surgery” and “bottom surgery”. The former refers to a set of procedures in which breast tissue is removed and the chest sculpted to resemble a more traditionally masculine form. As for the latter, this refers to a series of procedures in which the existing genitalia is altered to better have the

appearance and function of a natal penis. The two main bottom surgery options are metoidioplasty and phalloplasty, colloquially known as “meta” and “phallo”, respectively (Perovic and Djordjevic).

Medical intervention, especially surgery, is not accessible to a large number of transgender individuals. These interventions are often costly, time-intensive, and require access to adequate medical care. In the United States, bottom surgery often costs thousands of dollars as well as time, due to multi-stage procedures (Rashid and Tamimy). While the technology for both meta and phallo have been improving, there is a limited number of surgeons who practice either of these procedures and achieve desirable results. Their waitlists can be anywhere from months to years. Additionally, for those who do not wish to undergo a set of surgeries, options for limiting sexual dysphoria are limited to simple prosthetic devices, such as dildos and packers.

Societal Attitudes on Pleasure

There are cultural and historical tensions within this problem space, which traditionally uphold a narrative that pleasure is only accessible for those who are married, heterosexual, able-bodied, and seeking children (Tepper). In order to contextualize the modern-day attitudes and taboos around sexual pleasure, we must understand the political history of sex and development of social mores.

What forms of Sexual Pleasure are “Acceptable”?

Puranitical movements of the 19th century aimed to police the bodies of women and gay men, resulting in a culture of purity, shrouding sexuality in a black box (Rubin). These attitudes still persist. Espoused in legislation, media, and the consumerist world, this social framework creates a taboo against anyone it outcasts as a sexual deviant. Due to the ramifications of this taboo, there are forms of pleasure and bodies deemed “acceptable and natural”, while self-pleasure and the bodies of marginalized groups are not (Rubin). For those who are marginalized, there often exists an expectation to adhere to “normal” forms of pleasure and to strive for traditional sexuality in order to feel “complete” (Tepper). Any behavior that did not involve marriage, love, or reproduction in a heteosexual context was (and still is) viewed as destructive to the societal fabric.

Challenging the Status Quo

This particular taboo has been challenged at many different times, by many different groups. Second wave feminism, in particular, produced a great deal of literature and discussion surrounding the topic. These were often authored by one of the following: feminists who felt that the use of sex toys conflicted with the empowerment of women, those who embraced female pleasure but had hang-ups in introducing an inanimate technological device into the bedroom, or women who held concerns about how a sex toy might change their relationships (Lieberman). It should also be noted, that a great deal of this movement excluded discussions of a woman's pleasure outside of sex with a male partner. One such debate centered on whether the phallic appearance of a dildo meant that the toy was degrading to women (Williams). Lieberman goes on to explain that this debate also took into account the capitalistic element of the dildo as a product and if its consumption was ethical, citing the fact that Eve's Garden chose not to carry dildos initially due to all these controversies.

Other schools of thought labeled sex toys as dangerous, because they liberated women to explore their own bodies. The dildo is simultaneously both oppressing and liberating, within different contexts and for different people; to simply label the artifact as "good" or bad" would be to disregard the complexity of sexuality and pleasure (Das). Sex toys, such as the dildo, have use in different contexts to people of differing sexualities; dildos are used by couples, older patients with pain, people who are averse to partners, etc.

The Value of Sexual Pleasure

Regardless of this taboo, one's sexuality is a crucial component when considering quality of life for most people, whether cisgender or transgender (Holmberg et al.). While the dildo has historical roots as a medical apparatus and something to be hidden in nightstands, sex toys are beginning to be portrayed more favorably today. Part of this dialogue has demonstrated that there is value in maintaining sexual well-being and personal exploration. However, discussions around sexual health must involve social, psychological, and medical considerations; the expression of sexual identity is a manifestation of these facets (Jannini et al.).

There are already medical problem spaces that utilize sex toys as an intervention point, when addressing psycological health; it has been shown that co-designing sex toys to accomodate adults with motor disabilities allows these individuals to exercise greater agency

and comfort in their sexual and mental health (Morales et al.). The same can be done for alleviating dysphoria in transgender individuals. As such, there are researchers who assert that a shift in the cultural reading of the dildo as a taboo object into an empowering tool is already underway (Bardzell and Bardzell).

Sex toys are also incorporating modern technologies at an increasing rate, leading to specific branches of designer sex toys, such as “teledildonics” (Döring and Pöschl). Under sex positive frameworks, the designer dildo represents a tool for education and embracing one’s own body, both as a tool for sexual wellness and the ultimate HCI challenge.

Design Narrative

In stepping outside the bounds of traditional gender norms, transgender individuals often find themselves alienated from societal expectations of beauty, traditional narratives about relationships, and avenues for sexual pleasure. The main goal of this project was to make sex toys physically, emotionally, and financially accessible—as an intervention point for dysphoria commonly experienced by transmasculine individuals.

Final Design

A great deal of sexual dysphoria is centered around dissatisfaction with the appearance, function, and sensation of one's genitalia, but surgical interventions are either financially inaccessible or pose too many concerns. The alternative is the use of silicone prosthetics, which can mimic an affirming form but not function or sensation to the degree that most individuals would desire. The majority of these types of sex toys are overly simple and, without the ability to transmit sensation, do not allow a user to comfortably feel it as an extension of their own body. While there are some robotic or otherwise technologically advanced concepts in development, these tend to be extremely costly and complex to use (Jay).

The following design aims to fill in a niche between surgical intervention and basic sex toys that are too simple to address dysphoria. This builds upon a series of concepts, which will be outlined within the Design Process section, using stakeholder feedback from a large online survey.

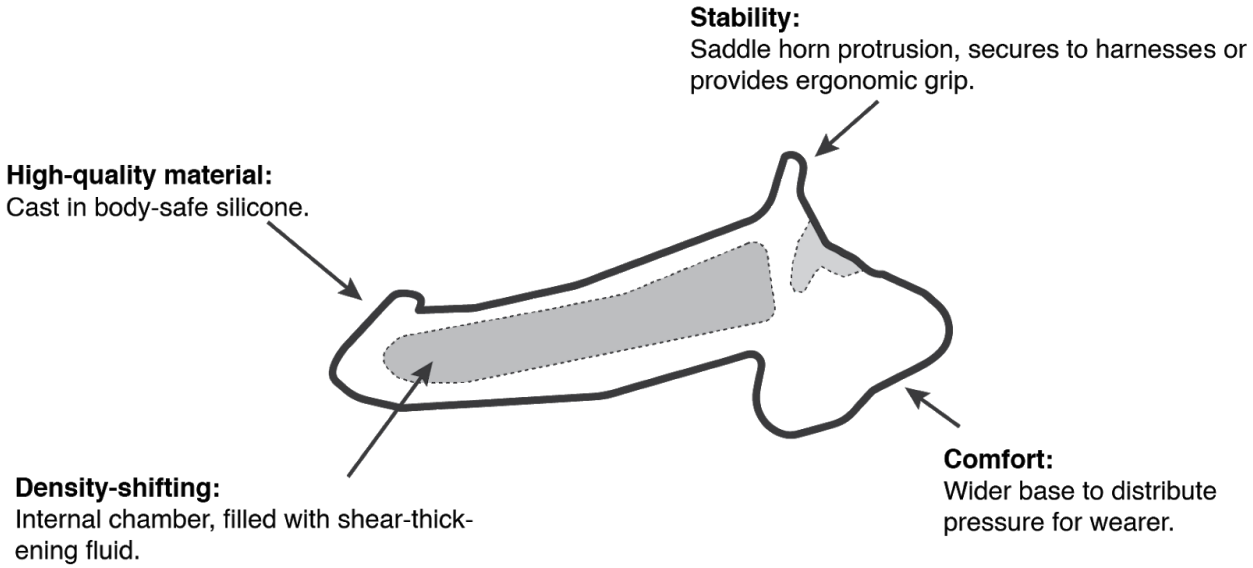

Figure 1 - Proposed design for a density-shifting dildo.

The main feature of this dildo is its internal chamber, which is manufactured by injection-molding silicone around a spacer. This spacer can then be removed during fabrication and the reservoir can be filled with a shear-thickening fluid, before being fully encased in the external silicone. This produces a core that shifts from liquid to solid when exposed to physical stimulus, mimicking the behavior of a natal penis.

Another key consideration addressed in this design is the need for physical and emotional comfort. As such, the base of this dildo design is wider (but thinner) and curves around the wearer's pelvis to comfortably absorb excess shock when penetrating a partner. Part of this base extends upwards to resemble a saddle horn-like structure, which is designed to interface with a harness to secure the dildo and prevent rotation about its major axis, which can often occur in traditional strap-ons. It should be noted that this protrusion is also meant to provide additional stimulation for the receiving partner.

A great deal of stakeholder feedback also emphasized the need for wearer stimulation in a pleasurable manner; the clitoral growth that most individuals undergo on HRT results in sensitivity and pinch points when transmasculine folks use strap-ons typically designed for cisgender women. To alleviate this discomfort and provide pleasure, the inside of the dildo's base is hollow and shaped in a way that can offer pleasurable sensation for individuals with bottom growth.

Currently, this is a materials approach that has not been explored by any other product on the market, so exact manufacturing costs are unknown. However, the ability for a dildo to alternate between solid-liquid states, without utilizing complex and expensive technologies, is incredibly unique as well.

Context within the Problem Space

Designing an artifact as intimate and personal as a sex toy presents a unique set of challenges. One of these is the awkwardness in speaking about one's needs in the bedroom under traditional interview settings. Another particular hurdle is how different anatomy can be for transgender individuals. This can be due to the fact that one's anatomy is constantly changing during the initial parts of hormonal transition, surgical interventions, or simply the fact that there is an incredible amount of diversity in body types. This particular challenge is compounded by the fact that there is a lack of data and representation of transgender bodies in mainstream studies. For example, while there are a number of open-source scans, diagrams, and 3D models of cisgender genitalia and bodies, this is not true for their transgender counterparts. In order to capture these many different personal experiences, I have utilized an anonymous survey to collect this stakeholder feedback and to co-design with the community.

Social Responsibility

As a designer (and someone privileged with education and resources) I believe that I have a responsibility to include the voices of those who've been historically underserved. When working with the LGBTQ+ community, this meant acknowledging the work of activists and researchers who came before me as well as maintaining anonymity when possible, to avoid outing any individual. Due to the personal and intimate nature of this project, efforts have been taken to protect the identities of those who volunteered their time and experiences through a digital barrier, the online survey (see Appendix).

Engineering the Anthropocene

Much of the literature in this space focuses on the ways in which operating under a capitalist society has negatively altered the environment on Earth (Raymond). As such, reparative readings and social justice movements look towards marginalized communities, many of which have embedded knowledge and practices for alternative ways of living

(Watson). While I did not intend to address the ecological ramifications of the Anthropocene with this project, I aimed to incorporate the grassroots methodologies espoused by these environmental activists.

Design Process

This section will outline the development of the project, through the initial ideation, problem space exploration and data collection, design iteration, and development of the final design. I will also discuss project challenges and lessons learned over the course of this thesis.

Ideation

While there are many types of sex toys, my initial instinct was to focus on designing a more affirming dildo for FTM and NB folks. Currently, the majority of dildos and strap-ons are aimed at users who are cisgender women; these products are often gendered in their marketing, design, and color palettes. In order to alleviate dysphoria and better serve the transgender community, this product should provide stimulation for the wearer, meet aesthetic preferences, be easy to use, and affirm one's gender in sexual contexts.

The first of these concepts (see below) was a dildo that could simulate the appearance and sensation of a penile erection.

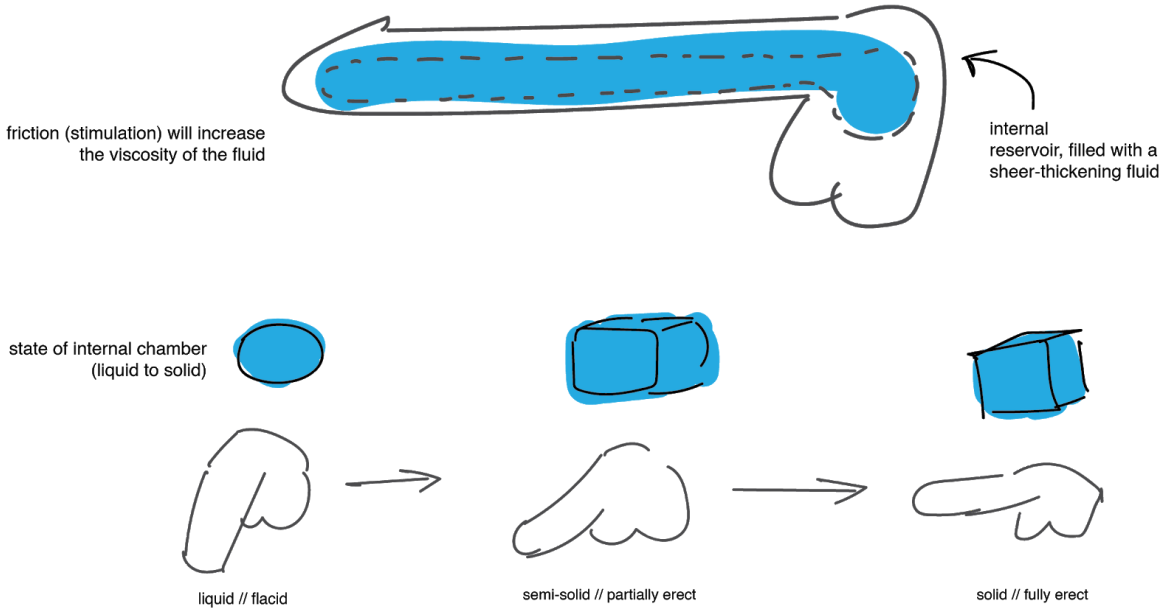

Figure 2 - Design sketches for a density shifting dildo.

Mechanically, it was designed to do so with an internal reservoir filled with a shear-thickening fluid. Since materials in this classification become more viscous as stress is applied, the toy would become more solid as stimulus intensified. As a proof of concept for this mechanism, a small-scale model was developed using common household materials. Pictured in Figure 3 is a thick rubber glove, filled with a non-Newtonian fluid called “oobleck”, which exhibits shear-thickening properties. Alongside the glove is a silicone “stroker” toy, which can be penetrated to provide stimulation.

Figure 3 - Proof of concept and materials selection.

When the glove is left at rest, the fluid is fully liquid and the glove is easily manipulated. As one of its digits is inserted into the toy, the oobleck begins to solidify until the glove is no longer malleable. Like a vast number of other sex toys, this product would have to be made of body safe silicone and the internal structure would have to be enclosed in such a way that the fluid would be rendered inert.

This toy could be worn with traditional strap-on harnesses, but aesthetically did little to challenge or change the typical aesthetic of sex toys. Part of the rationale for maintaining this was the fact that nearly all packers and FTM specific prosthetics tend to be realistic; perhaps this was key to alleviating dysphoria. It should also be noted that this early concept was developed prior to the majority of stakeholder feedback. Therefore, this initial concept is grounded primarily in personal experience and intuition.

Survey Distribution

While I have personal goals and stakes for this problem space, I wanted to include the opinions and voices of other queer folks in the design process. In order to capture the experiences of other members of the target demographic, a survey was distributed on the following social media channels:

- The TRANS* Group (an RPI Facebook/Discord group of current and graduated students who identify as transgender)
- NY Gendernet (Discord server that connects trans folks from all over NYS)
- Queer Asian Intersections (Facebook group for folks who are queer and Asian)
- SQuATTING (a Facebook support group for Asians who self-identify as Trans, Intersex, Gender Non-Conforming, Non-binary etc.)
- r/ftm
- r/asktransgender
- r/SexToys

This approach proved to be incredibly effective, with over a hundred responses over the course of two weeks. Through comments and feedback, I was able to isolate common experiences, pain points, and needs. With these, I developed a list of key features, which will be discussed further in this section.

Survey Demographics

The single largest age group represented in this data are college-aged, young adults; nearly two-thirds of the responses came from individuals who are 18 to 24 years old. Additionally, the overwhelming majority (96.7%) of respondents are between the ages of 18 to 34. This skew in age groups could be attributed to both the demographics reached via online canvassing or the amount of sexual activity typically associated with age groups.

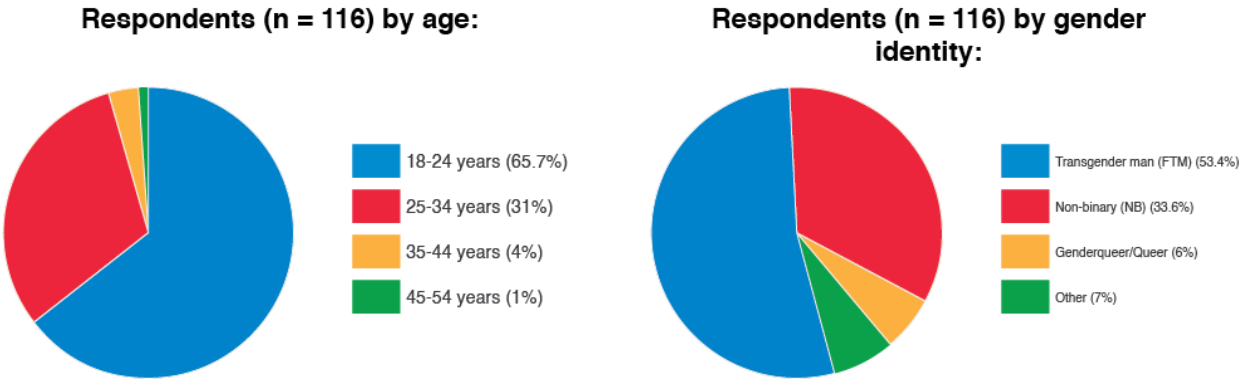

Figure 4 - Respondents by age group and gender identity.

It should also be noted that 53.4% of respondents identified as FTM transgender men and 33.6% identified as non-binary. The needs and experiences of these groups overlap, but are not always necessarily the same.

Stakeholder Feedback

The following are the categories which respondents were asked to their rank preferences for, on a 7-point Likert scale:

- Texture
- Density
- Ability to transmit sensation
- Vibration
- Ability to transition from flacid to erect state
- Aesthetics

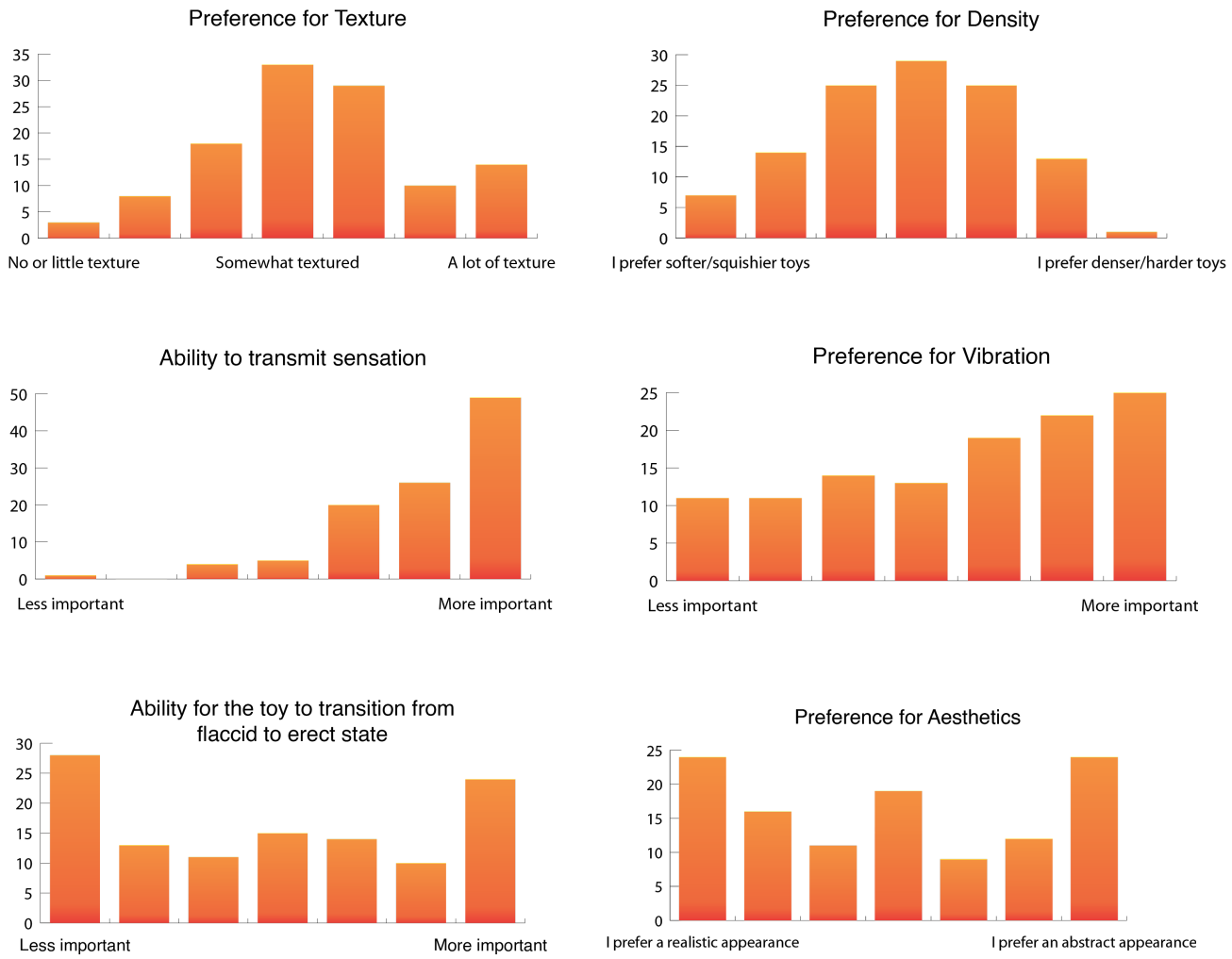

Figure 5 - Stakeholder responses.

Based on feedback and the Likert scale questions, these are the key features, listed in order of importance.

- Ability to transmit sensation and give wearer pleasure, which few strap-ons currently provide.
- Density on the softer side, as more solid toys can be uncomfortable for both the wearer and the receiving party.
- Medium amount of texture
- Ability to “rumble” vibrate
- Ability to transition from flaccid to erect
- Realistic appearance, notable preference for uncircumcised, especially from respondents outside the US.

Additional Findings from the Survey Data

- Of the possible medical interventions, the most popular was HRT, which 76.7% of respondents have pursued or plan to pursue. Those who did not express interest were primarily those who identify as NB.
- The second most popular medical intervention was top surgery, which 69% of respondents have pursued or plan to pursue. Additionally, 44% expressed the same sentiment for hysterectomy.[1]
- The majority of transmasculine people within this data set have not or do not intend to pursue options for bottom surgery. Only 10.3% of respondents planned to pursue meta and 18.1% other respondents planned to pursue phallo.[2]
- When pleasuring a partner, 75.2% of respondents penetrate their partners.
    - In contrast, the places where respondents liked receiving pleasure differed immensely. Nearly all of this was due to dysphoria, which renders certain parts of one's body uncomfortable to engage with.
- The most important feature of a toy used for self-pleasure is vibration, which 79.1% of respondents emphasized.

Additionally, comments from stakeholders provided tangible evidence for a number of my intuitions/assumptions in the initial concept.

- **Lack of options:** While many individuals were able to find toys that addressed their sexual needs, this involved trial and error. Most respondents dedicated money and time to purchasing and trying out different types of toys because these were often unsatisfactory; there are few products that fit the niche of alleviating sexual dysphoria in an effective manner.
- **Aesthetics:** All comments regarding unrealistic dildo colors stressed that the design must not use colors that are traditionally female-coded, such as pink or purple.
- **Impact of gender identity:** The majority of NB respondents expressed a desire for prosthetics that are non-realistic, fantasy, or otherwise abstract. Many of them cited the fact that a more realistic dildo can set off dysphoria or often ventures into "uncanny valley".

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[1] This data point suggests that there may not necessarily be an aversion to surgical options.
[2] In comparison to the two other types of GAS, there was a notable decrease in consideration for bottom surgery.

- In contrast, the majority of binary FTM respondents prefer realistic dildos and packers, but a number have noted that current products have limited skin tone options.

Interview with Sarah Jayne Kinney

While the online survey allowed me to gather feedback from the target audience, I also wanted to better understand the problem space through the eyes of someone who worked in the sex toy industry. As such, I reached out to Sarah Jayne Kinney, co-founder of *Unbound*, which is one such company that caters to an audience that's mostly neither cisgender nor heterosexual. When it was founded in 2012, the company's product was a subscription box, curated to include sex toys "that anyone could use" (Sarah Jayne Kinney). As they began to pivot towards creating original products, there were several key considerations.

Sustainable packaging and an open dialogue with marginalized communities were addressed, informing my design choices and materials selection. Additionally, one of the main constraints in the *Unbound* design process is financial accessibility; Sarah Jayne highlighted the fact that all of the products were below a $100 price point. It should be noted that this is a departure from the typical cost of transgender prosthetics, which typically cost hundreds of dollars. While *Unbound*'s products still follow traditional sex toy design and are not trans-specific prosthetics, this serves as a reminder that an approachable cost is both important and feasible. As such, this design constraint is one that I have adopted into this project as well.

Iteration

While the large number of respondents affirmed the need for this type of work, the sheer breadth of feedback made it difficult to narrow down one specific set of needs to address. Since sexual pleasure is so intimate and personally specific, comments were very polarized. For example, when asked about likes/dislikes regarding type of toy, one respondent commented:

"Definitely enjoy dildos, both for myself and penetrating a partner. Mostly because it feels super gender affirming to be about to penetrate someone with something attached to your own body. It's like an extension of yourself."

On the other hand, another individual expressed a contrasting opinion that “dildos on me give me dysphoria”. It began evident that there was not a unified set of opinions and therefore, not a design that could adequately address the needs of everyone. As a designer, I made the choice to continue focusing on designing a better dildo, since it was the type of toy that the majority of respondents preferred to use, both for themselves and partners.

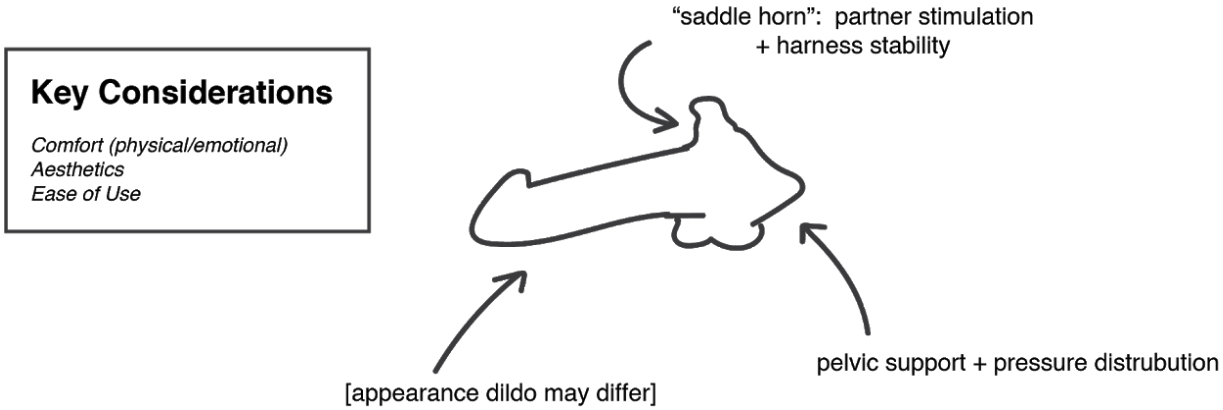

Figure 6 - Design for a more stable and comfortable dildo.

While this second concept would still feature the internal reservoir of shear-thickening fluid, the majority of new iteration lies in the external design. Since the primary consideration is comfort, the base of this design is wider and curves around the pelvis to better distribute pressure when penetrating a partner. The base also features a saddle horn-like protrusion, which can act as an anchor point when securing a dildo to a harness. This particular consideration developed from user feedback that cited a lack of stability and high slip-rates in traditional o-ring designs. Additionally, the shape would also allow a user to better grip the dildo if using the toy on themselves.

Figure 7 - The spectrum of morphological preferences.

Moving forwards, I chose to pursue a prototype design that was more realistic in nature, due to the preferences expressed in the survey and the fact that a simple shape is easier to manufacture. Using carving wax, I crafted a hollow dildo in order to cast silicone replicas, which could then be filled with a shear-thickening fluid. Based on stakeholder feedback, the model is designed to resemble a natal penis that has been uncircumcised, which is more in line with modern and international cultural practice.

Figure 8 - Initial stages of the wax carving (left) and the finished wax model (right).

After the mold was made and a silicone model was produced, I realized how daunting this process would be to a user with no technical background. Initially, I had considered a design which would allow a user to buy a kit with this dildo semi-finished, so that they could take a mold of their own anatomy and finish the base with a custom cast. This would be inconvenient and overly complex to the point that it would impinge upon the goal of increasing ease of use, so this component to the product's user experience was removed.

Next Steps

A number of the comments on the survey were messages of gratitude that this problem space was finally being addressed, affirming my intuition that this project holds a great deal of value and potential for the transgender community. As such, one of the effective means of developing this project into a product would involve partnering with an existing sex toy company, such as *Unbound*. Another approach would be securing a patent and working towards developing my own company, which is logistically and financially more complex. As Sarah Jayne mentioned to me, while there is less taboo around sexual pleasure as a whole, many individuals and organizations are still hesitant to support a company that operates outside the mainstream. Yet, given the speed at which transgender stakeholders responded to my survey and the support espoused within their comments, there is likely grassroots funding available for this type of endeavor.

The materials needed for production of the final design are commercially available, but can often be costly. I would have to secure reliable access to high quality silicone and the equipment needed to reliably cast or mold. I would also like to acknowledge that silicone is not a sustainable material, as it is a non-biodegradable silica derivative. However, there are very few materials that are body-safe, inert, and easily manufacturable which can be used in a sexual setting.

Conclusion

Sexual pleasure is a core component of physical, mental, and emotional stability. However, for transgender individuals, the bedroom may be an incredibly uncomfortable space. We can neither adhere to our own expectations of our bodies nor the societal narrative around what sex should be. This paper highlights the importance of creating a space where discussions around such taboo and intimate subject matter can occur. As individuals who often experience sexuality in ways that challenge an inaccurate paradigm, transgender individuals are a wealth of knowledge when it comes to unboxing the black box of how gender influences sexuality and pleasure.

While the general public is delving into this uncharted territory of sexual freedom, with designer sex toys and more avenues of pleasure, we are often alienated from these options, traditional relationships, and therefore, a key element in maintaining our well-being: love and comfort in one's sexuality. Current medical interventions are often unattainable due to cost, not pursued because of unsatisfactory results, or present complications that some find too risky. I recognize that, as both a designer and a transgender individual, I exist at a specific set of situated knowledges; this is a problem space that I have the tools to address. By working off of my personal experiences and intuitions, I have been able to develop a unique design for a sex toy, using feedback from other transgender stakeholders. This product is specifically created to address the comfort, to improve sexual satisfaction, and to alleviate dysphoria in the bedroom.

While this product cannot fully replace the more permanent and intimate results of GAS, it is intended to bridge the gap between costly procedures and traditional prosthetics, which have not truly been improved since the introduction of silicone.

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Appendix

Designing better sex toys for the FTM trans and AFAB NB communities.

This survey is designed to gauge unmet needs and desires for those who are AFAB and identify as transgender, genderqueer, gender non-conforming, etc.

The responses on this form are anonymous and will never be published. Feedback will only be used in the scope of this project and developing a series of prototypes (making new sex toys that directly serve the trans/gnc bodies).

You must be 18 years of age or older to fill out this survey. By proceeding, you are confirming that you meet these criteria.
* Required

Demographic Information

This information will not be shared or used to identify you. This is being collected in order to contextualize your responses and feedback.

1. What is your age? *

Mark only one oval.

- 18-24 years
- 25-34 years
- 35-44 years
- 45-54 years
- 55+

2. Which label (if any) best describes your gender? *

Mark only one oval.

- Transgender man (FTM)
- Non-binary (NB)
- Queer
- Genderqueer
- Gender non-conforming (GNC)
- Other: _____

3. Which (if any) of the following gender-affirming options have you pursued or plan to pursue? *

Check all that apply.

- [ ] None
- [ ] Testosterone/Hormone Replacement Therapy (HRT)
- [ ] Top Surgery (any procedure within this category)
- [ ] Hysterectomy (any procedure within this category)
- [ ] Metoidioplasty (Meta)
- [ ] Phalloplasty (Phallo)

Other: [ ] _____

4. What parts of your body do you like to engage with when pleasuring yourself? *

_____

5. What parts of your body do you like to use when pleasuring a partner? Ex: Hands, mouth, etc. *

_____
_____
_____
_____
_____

6. Where do you like being pleasured? This can include erogenous zones anywhere on the body. *

_____

7. Are there parts of your body that you are uncomfortable engaging with due to dysphoria or any other reason? *

_____
_____
_____
_____
_____

Toy usage in the bedroom.

This section is meant to better understand how folks like using their toys.

8. When I masturbate, I like toys that involve...

Check all that apply.

- [ ] Vibration
- [ ] Penetration
- [ ] Suction
- [ ] Stroking/grinding

Other: [ ] _____

9. When I'm using toys with a partner, I like toys that involve...

Check all that apply.

- [ ] Vibration
- [ ] Penetration (for myself)
- [ ] Penetration (for my partner)
- [ ] Suction
- [ ] Stroking/grinding

Other: [ ] _____

Experiences with toys.

This next set of questions is to better understand what types of toys would you be most interested in developing.

10. What type(s) of toy do you already own/use? *

Check all that apply.

- [ ] None
- [ ] Packers
- [ ] Dildo (for personal play)
- [ ] Dildo (for penetrating a partner)
- [ ] Plugs
- [ ] Bullet Vibrator
- [ ] Wand Vibrator
- [ ] G-spot/Clitoral Vibrator
- [ ] Stroker/Masturbator

Other: [ ] _____

11. If you have owned/used any of these toys, which did you like the most? Why? Feel free to include links to your favorite products as examples. *

12. Of the toys you have not used, which type(s) of toy might you be interested in trying? *

Check all that apply.

- [ ] None
- [ ] Packers
- [ ] Dildo (for personal play)
- [ ] Dildo (for penetrating a partner)
- [ ] Plugs
- [ ] Bullet Vibrator
- [ ] Wand Vibrator
- [ ] G-spot/Clitoral Vibrator
- [ ] Stroker/Masturbator

Other: [ ] _____

13. Have there been specific toys or types of toys that you've experienced frustration with? How so? *

14. Of the previous list of toys, which would you most like to see redesigned or improved? *

Preference for features/functionalities in toys.

Please rank how you value the following features.

15. Textures

Mark only one oval.

	1	2	3	4	5	6	7	
I prefer toys that have little texture								I prefer toys that have a lot of texture

16. Density

Mark only one oval.

	1	2	3	4	5	6	7	
I prefer softer/squishier toys								I prefer harder/denser toys

17. The ability to transmit physical sensation (such as grinding, suction, or receiving/giving penetration)

Mark only one oval.

	1	2	3	4	5	6	7	
Less important								More important

18. Vibration

Mark only one oval.

	1	2	3	4	5	6	7	
Less important								More important

19. The ability for the toy to transition from flaccid state to erect state

Mark only one oval.

	1	2	3	4	5	6	7	
Less important								More important

20. Aesthetics

Mark only one oval.

	1	2	3	4	5	6	7	
I prefer toys that appear realistic	◯	◯	◯	◯	◯	◯	◯	I prefer toys that appear colorful/abstract

21. Please describe your preferences in terms of a toy's aesthetics, color palettes, or shapes. Feel free to name specific toys you like.

22. Do you have any additional insights/comments?

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